

Amirmehdi “Amir” Moghadamfarid

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Research Profile

My research interests lie in AI/ML-assisted security, privacy, and safety for connected systems, especially IoT, embedded, edge, and distributed environments. My work combines systems-building, security analysis, trusted hardware, distributed storage, and emerging LLM-based methods for detecting and mitigating malicious behavior in real-world connected infrastructures.

Education

Michigan Technological University

Ph.D. Student in Computer Science; M.S. expected Fall 2026

GPA: 3.87/4.00

Houghton, MI, USA

Dec. 2024 – Present

Islamic Azad University

B.S. in Computer Engineering

GPA: 3.2/4.00

Tehran, Iran

Sep. 2016 – Jul. 2021

Publications

Machine Learning for Parkinson’s Disease: A Comprehensive Review of Datasets, Algorithms, and Challenges

Amirmehdi Moghadamfarid, [Co-authors]

[Journal/Conference Name]. Published, Jul. 2025.

Fog Computing Approaches in IoT-Enabled Smart Cities

Amirmehdi Moghadamfarid, [Co-authors]

[Journal/Conference Name]. Published, 2023.

Submitted and In-Preparation Manuscripts

Secure Decentralized Cloud Storage Assisted by Trusted Hardware and Erasure Coding

Amirmehdi Moghadamfarid, Josh Dafoe, Bo Chen

Computers & Security, Elsevier. Submitted, Aug. 2025.

File-System-Friendly Audit-Less Decentralized Cloud Storage with TEE, FTL, and Network Coding

Amirmehdi Moghadamfarid, [Co-authors]

Manuscript in preparation. In preparation.

Safeguard: LLM-Based Detection of Malicious Agent Behavior in IoT Environments

Amirmehdi Moghadamfarid, [Co-authors]

Manuscript in preparation / ongoing project. In preparation.

Teaching LLM Attacks to K–12 Students: Security Education for Emerging AI Threats

Amirmehdi Moghadamfarid, [Co-authors]

Manuscript in preparation / ongoing project. In preparation.

Research Experience

Graduate Research Assistant

Cybersecurity Lab, Michigan Technological University

Dec. 2024 – Present

Houghton, MI, USA

Project: Secure Decentralized Cloud Storage Assisted by Trusted Hardware and Erasure Coding

- Developed a C-based research prototype for secure decentralized cloud storage using erasure coding and cryptographic protection mechanisms.
- Implemented and evaluated Reed–Solomon erasure coding, AES-based protection, and trusted-execution-assisted design components.
- Evaluated throughput, scalability, fault tolerance, and resilience under adversarial and real-world system assumptions.
- Studied security and reliability tradeoffs in decentralized storage systems using trusted execution environments such as Intel SGX and related TEE models.

Project: File-System-Friendly Audit-Less Decentralized Storage

- Investigating storage-layer mechanisms for recognizing and protecting authenticated writes in file-system-friendly decentralized storage.
- Studying the interaction between file systems, flash translation layer behavior, trusted hardware, and network-coding-based recovery.
- Exploring audit-less integrity mechanisms that reduce verification overhead while preserving security and retrievability guarantees.

Project: Decentralized Finance Security and Reliability

- Explored security, reliability, and scalability aspects of decentralized finance systems, including the Stellar Network.

Research Assistant

IoT Lab

Jul. 2020 – Feb. 2024

Tehran, Iran

Project: Fog Computing for IoT-Enabled Smart Cities

- Conducted a comprehensive study of fog-computing approaches for latency-sensitive and security-critical smart city applications.
- Analyzed fog-based architectures for real-time and resource-constrained IoT environments, including healthcare monitoring and smart manufacturing.

Project: Machine Learning for Parkinson's Disease

- Surveyed machine learning approaches for Parkinson's disease analysis across acoustic data, biomarkers, medical imaging, movement data, and multimodal datasets.
- Categorized datasets, algorithms, and research challenges in ML-assisted healthcare analysis.

Industry Experience

Backend Developer

Parchenegar.com

Mar. 2019 – Jan. 2023

Tehran, Iran

- Designed and maintained server-side logic and APIs for multi-branch warehouse management systems.
- Developed backend functionality using PHP and Laravel.

Teaching and Mentoring

Teaching Assistant

Michigan Technological University

[Semester, Year]

Houghton, MI, USA

- Supported instruction, office hours, grading, and student debugging for [Course Name].
- Assisted students with programming, systems, security, or computer science concepts.

Volunteer Tutor

Computer Association, Islamic Azad University

Sep. 2018 – Feb. 2020

Tehran, Iran

- Taught Linux LPIC-1 fundamentals to undergraduate students.

Leadership and Service

CS Department Representative

Graduate Student Government, Michigan Technological University

Sep. 2025 – Present

Houghton, MI, USA

- Represented graduate student interests in the Computer Science department through Graduate Student Government activities.

Technical Skills

Programming	Python, C/C++, Bash, LaTeX
Systems	Linux, distributed systems, storage systems, file systems, flash translation layer (FTL), embedded systems, Docker, Git
Security	Cryptography, trusted hardware, trusted execution environments, attestation, proof of retrievability, proof of storage, Merkle trees
AI/ML	LLMs, machine-learning-based detection, scikit-learn, pandas, data analysis
Research Areas	IoT security, embedded systems security, edge systems, decentralized storage, network coding, erasure coding, systems security
Languages	Persian/Farsi (native), English (fluent), Turkish (conversational)

Selected Academic Strengths

- GRE: Quantitative 170, Verbal 168, Analytical Writing 4.0.
- Graduate GPA: 3.87/4.00.
- Research background spanning IoT, machine learning, trusted hardware, decentralized storage, LLM security, and systems security.

References

Available upon request.